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**Improving Frontier Exhaustion
Recovery in Decentralized Multi-
UAV Exploration
BACHELOR THESIS**

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ABSTRACT

Keywords: frontier exhaustion, multi-UAV systems, decentralized exploration, GPS-denied environment, topological graph.

In GPS-denied indoor and underground settings, the exploration of multi-UAVs is decentralized and limited to communication constraints as well as SLAM drift that leads to agent coordination imbalances. It examines frontier exhaustion within the GVP-MREP base system where the local Voronoi cell of a UAV exhausts all possible frontier targets as other agents proceed to explore the system, leading to longer idle times and less efficient coverage.

It aims at defining this failure mode and suggesting a specific improvement of recovery to be included in the current graph-based coordination workflow. The Improving Frontier Exhaustion Recovery (IFER) algorithm identifies exhaustion by detecting silence in the trajectory and Dijkstra-based EROI reassignment over the shared MR-DTG without altering the underlying GVP-MREP stack. Three simulation environments (testcity, maze3, and maze4) with 30 baseline and recovery configurations were used in experiments.

Findings indicate that the recovery mechanism is effective to restore idle UAVs across all the environments tested. Maze4 showed the greatest reduction in the time of simulation, whereas testcity authenticated restoration of global planning activity which was missing in the base. The results of maze3 show that the geometry of corridors is still a constraining factor.

It is limited to only simulation-based validation, sensitivity to extreme SLAM drift, and fixed UAV team size. The validation should be expanded to physical UAVs, larger teams, and recovery strategies based on learning in the future.

Thesis volume: ~64 pages, 16 tables, 11 figures, 2 appendices, 25 references.

ANOTĀCIJA

Atslēgvārdi: robežu izsīkums, vairāku bezpilota lidaparātu sistēmas, decentralizēta izpēte, GPS liegta vide, topoloģiskais grafs.

GPS liegtā iekštelpu un pazemes vidē vairāku bezpilota lidaparātu izpēte ir decentralizēta un ierobežota ar komunikācijas ierobežojumiem, kā arī SLAM novirzi, kas noved pie aģentu koordinācijas nelīdzsvarotības. Šajā darbā tiek pētīta robežu izsīkums GVP-MREP bāzes sistēmā, kur bezpilota lidaparāta lokālā Voronoi šūna izsmel visus iespējamus robežu mērķus, kamēr citi aģenti turpina izpētīt sistēmu, kā rezultātā ilgāks dīkstāves laiks un mazāk efektīva pārklājuma nodrošināšana.

Darba mērķis ir definēt šo kļūmes režīmu un ieteikt konkrētu atkopšanas uzlabojumu, kas jāiekļauj pašreizējā uz grafiem balstītajā koordinācijas darbplūsmā. Improving Frontier Exhaustion Recovery (IFER) algoritms identificē izsīkumu, nosakot klusumu trajektorijā un uz Deikstras balstītu EROI pārdali koplietotajā MR-DTG, nemainot pamatā esošo GVP-MREP steku. Eksperimentos tika izmantotas trīs simulācijas vides (testcity, maze3 un maze4) ar 30 bāzes un atkopšanas konfigurācijām.

Pētījuma rezultāti liecina, ka atkopšanas mehānisms ir efektīvs, lai atjaunotu dīkstāvē esošos bezpilota lidaparātus visās testētajās vidēs. Maze4 uzrādīja vislielāko simulācijas laika samazinājumu, savukārt testcity autentificēja globālās plānošanas aktivitātes atjaunošanu, kuras nebija bāzē. Maze3 rezultāti liecina, ka koridoru ģeometrija joprojām ir ierobežojošs faktors.

Tā ir ierobežota tikai ar simulācijā balstītu validāciju, jutību pret ārkārtēju SLAM nobīdi un fiksētu bezpilota lidaparātu komandas lielumu. Validācija nākotnē būtu jāpaplašina, iekļaujot fiziskās bezpilota lidaparātus, lielākas komandas un atgūšanas stratēģijas, kas balstītas uz mācīšanos.

Disertācijas apjoms: ~64 lappuses, 16 tabulas, 11 attēli, 2 pielikumi, 25 atsauces.

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LIST OF ABBREVIATIONS

Abbreviation	Full Name
ACHORD	Communication-Aware Multi-Robot Coordination with Intermittent Connectivity
DARPA	Defense Advanced Research Projects Agency
EROI	Explorable Region of Interest
FUEL	Fast UAV Exploration Using Incremental Frontier Structure
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GVP	Graph Voronoi Partition
GVP-MREP	Graph Voronoi Partition Multi-Robot Exploration Planning
IFER	Improving Frontier Exhaustion Recovery
IR2	Implicit Rendezvous for Robotic Exploration Teams
LiDAR	Light Detection and Ranging
MR-DTG	Multi-Robot Dynamic Topological Graph
RACER	Rapid Collaborative Exploration with a Decentralized Multi-UAV System
RGB-D	Red Green Blue – Depth
ROS	Robot Operating System
SLAM	Simultaneous Localization and Mapping
SubT	Subterranean (DARPA Subterranean Challenge)
UAV	Unmanned Aerial Vehicle

INTRODUCTION

Indoor and subterranean environments such as collapsed buildings, tunnels, mines, and underground infrastructure represent some of the most challenging spaces for post-disaster assessment and inspection. These environments are often dangerous, inaccessible, or time-critical for human responders, which has driven increasing interest in autonomous robotic exploration solutions over the last decade [1]. Small unmanned aerial vehicles (UAVs) are particularly attractive in this context because they can traverse confined three-dimensional spaces, overcome uneven terrain, and reach locations that are difficult or impossible for ground robots to access [1], [2].

A defining feature of indoor and subterranean missions is the absence or unreliability of global navigation satellite systems (GNSS/GPS). As a result, UAV's operating in such environments must rely entirely on onboard sensing and autonomy for navigation and mapping [3]. Recent surveys of GPS-denied UAV navigation emphasize that robust autonomy typically requires tightly coupled perception and planning pipelines, combining inertial sensing with LiDAR or vision-based mapping to estimate motion and reconstruct the environment online [3], [4].

Large-scale field deployments, most notably those conducted within the DARPA Subterranean (SubT) Challenge, have demonstrated both the potential and the limitations of UAV exploration in underground environments [4]. These experiments show that while aerial robots can provide rapid situational awareness, real subterranean environments include repetitive structures, low-texture surfaces, dust, smoke, and long corridors that significantly degrade localization accuracy and increase drift over time [2], [5]. Consequently, exploration decisions are made based on an evolving and partially observable map rather than a globally consistent representation of the environment [5].

Efficient exploration therefore requires not only accurate mapping but also effective decision-making under uncertainty. Modern exploration systems commonly use occupancy or voxel-based grid maps to represent free, occupied, and unknown space, as these representations support incremental updates and probabilistic reasoning [6], [7]. On top of such maps, frontier-based exploration remains a widely adopted strategy because it directly targets the boundary between known and unknown regions to maximize information gain [1], [7]. However, recent studies highlight that frontier-

based methods are sensitive to localization errors and can become inefficient when the map is inconsistent or frequently updated due to delayed corrections [6].

The exploration process can be significantly accelerated by deploying multiple UAVs simultaneously. Decentralized multi-UAV exploration has therefore become an active research topic, with recent work demonstrating improved coverage speed and robustness compared to single-robot systems [8]. In practice, each UAV maintains its own local map and selects actions based on local observations, as maintaining a continuously synchronized global map is often infeasible [9]. This decentralization introduces coordination challenges, as independent agents may select overlapping exploration targets or revisit already explored regions [8], [10].

Communication constraints further complicate multi-UAV exploration. Surveys of multi-robot communication report that indoor and subterranean radio propagation is characterized by limited bandwidth, variable latency, and packet loss, which restricts the frequency and size of information exchange between agents [11]. As a result, dense map sharing or frequent global synchronization cannot be assumed in realistic deployments [11], [12]. Recent coordination frameworks therefore emphasize communication-aware strategies that rely on compact information exchange and remain effective under intermittent connectivity [10], [12].

Recent work has shown that communication-efficient decentralized exploration can also be approached through compact graph-based coordination rather than only through dense map sharing or handcrafted spatial decomposition rules [10], [11], [12].

Among such methods, GVP-MREP is especially relevant because it combines a multi-robot dynamic topological graph with graph Voronoi partition to reduce communication overhead while preserving effective task allocation in multi-UAV exploration. This makes it a suitable baseline for studying decentralized exploration under communication constraints [13]. The goal of the bachelor thesis is to analyse frontier-exhaustion behaviour in decentralized multi-UAV exploration under communication constraints and to propose a targeted recovery improvement for the selected baseline system. The proposed recovery mechanism is referred to as the IFER (Improving Frontier Exhaustion Recovery) method and is designed to integrate directly into the existing GVP-MREP workflow.

The tasks of the bachelor thesis are the following:

- (1) Analyse the state of the art in decentralized multi-UAV exploration and frontier-based methods, identifying assumptions that weaken under realistic communication constraints.
- (2) Review representative exploration systems relevant to the selected problem, with emphasis on RACER, GVP-MREP, and related communication-aware approaches.
- (3) Explain the architecture of GVP-MREP and identify the mechanism through which frontier exhaustion can appear.
- (4) Define evaluation criteria for exploration activity, communication volume, and frontier-exhaustion severity and frequency.
- (5) Conduct a simulation-based baseline analysis using recorded GVP-MREP runs in the testcity, maze3, and maze4 environments.
- (6) Propose a targeted recovery improvement for frontier-exhaustion handling and describe how it can be integrated into the existing workflow.
- (7) Analyse the obtained results, discuss limitations, provide recommendations, and propose directions for further work.

1. DECENTRALIZED MULTI-UAV EXPLORATION IN GPS-DENIED ENVIRONMENTS

1.1 Problem formulation for decentralized multi-UAV exploration in GPS-denied environments

Subterranean and indoor environments are widely recognized as one of the most challenging domains for autonomous robotics due to unknown structure, limited sensing range, and the absence of global positioning [1], [2]. Surveys of subterranean exploration explicitly classify these environments as GPS-denied, requiring fully autonomous onboard navigation and mapping [1].

Subterranean environments challenge UAV autonomy due to GPS absence and complex geometry [1], [2]. Literature analysis focuses on decentralized exploration under communication limits [8], [9]. Recent surveys emphasize the need for robust coordination without global maps [3], [4]. Fig. 1.1 shows the scope of this literature review [1], [8].

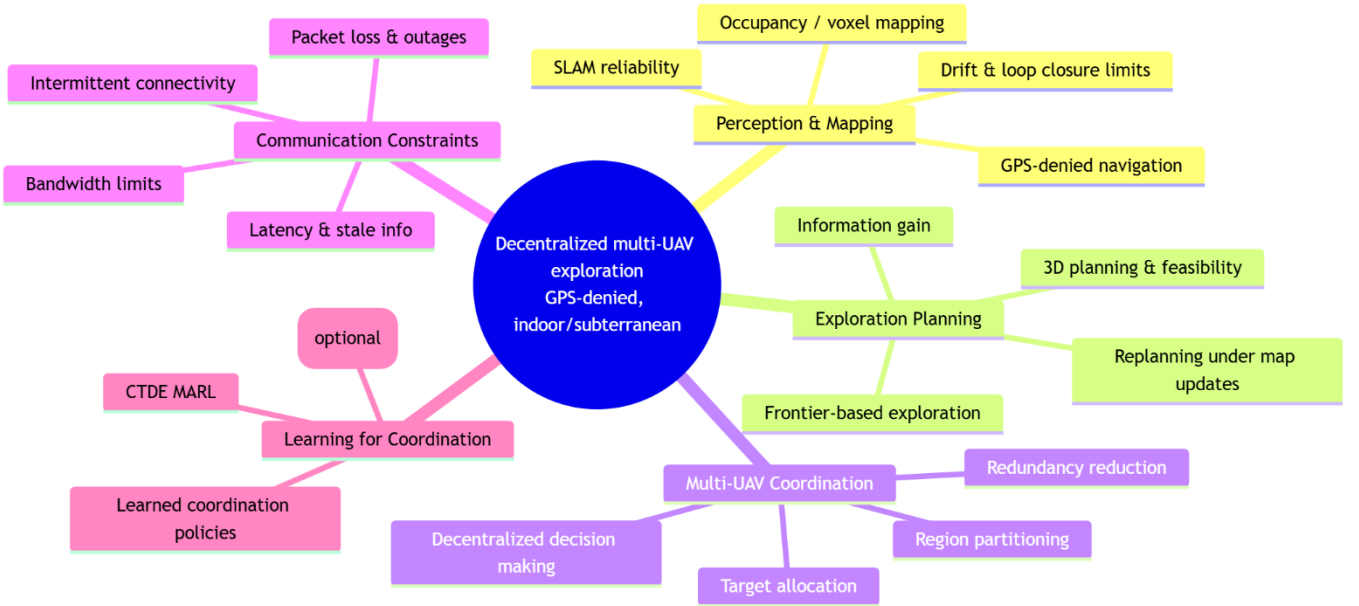


Fig. 1.1. Scope and structure of the literature analysis for decentralized multi-UAV exploration [1], [2], [4], [8], [9], [11]

The diagram covers frontier-based methods, decentralized coordination, and communication-aware systems. It positions GVP-MREP within the relevant research landscape [13]. This structure guides the baseline selection further. In such

environments, UAVs must estimate their motion using onboard sensors while simultaneously building a representation of the surrounding space. Reviews of GPS-denied UAV autonomy identify SLAM-based approaches as the dominant solution, typically combining inertial measurement units with LiDAR or visual sensing [3]. However, recent SubT-focused analyses show that underground SLAM performance degrades in long tunnels and repetitive geometries, where perceptual aliasing and delayed loop closures are common [4].

The consequences of SLAM drift extend beyond localization accuracy. Exploration planners rely on the current map estimate to select feasible and informative targets, so map inconsistency can lead to redundant exploration or unreachable goals [5]. Work on volumetric exploration under severe odometry drift demonstrates that planners must explicitly tolerate evolving maps rather than assuming global consistency [5].

In multi-UAV systems, the problem becomes inherently decentralized. Each UAV has access only to its local observations and map estimate, and global synchronization is rarely achievable in real time [7]. Studies on decentralized multi-UAV exploration emphasize that coordination must operate under partial observability, where agents cannot reliably know the state or intentions of other agents at decision time [7], [8].

Communication constraints further shape the problem formulation. Reviews of multi-robot communication in confined environments show that bandwidth limitations, latency, and packet loss significantly restrict the amount of information that can be exchanged. As a result, realistic formulations assume that only compact, high-level information such as target identifiers or summarized map regions can be shared intermittently [8], [10].

1.2 Frontier-based exploration

To support exploration in unknown environments, robots require a representation that can encode uncertainty and be updated incrementally. Occupancy grids and voxel-based maps remain widely used because they provide probabilistic representations of free and occupied space and support efficient collision checking and planning [5], [6]. Recent mapping frameworks focus on scalability and memory efficiency to enable real-time operation on UAV platforms with limited onboard resources [6].

Within such representations, frontier-based exploration is one of the most established strategies for autonomous mapping. Frontiers are defined as the boundary between known free space and unexplored regions, and navigating toward them is likely to reveal new information [1], [6]. Surveys on confined and subterranean exploration consistently identify frontier-based planners as a practical baseline due to their computational efficiency and simplicity [1].

Recent UAV-specific exploration systems extend classical frontier methods to three-dimensional environments by clustering frontier cells and scoring candidate regions based on expected information gain and travel cost [7]. For example, hierarchical frontier planners integrate global frontier selection with local trajectory optimization to enable fast replanning in unknown spaces [7].

However, literature on drift-aware exploration shows that frontier planners are sensitive to map inaccuracies. When loop closures occur or maps are corrected, previously identified frontiers may disappear or move, requiring planners to adapt dynamically [5]. This sensitivity becomes more pronounced in multi-UAV systems, where different agents may hold inconsistent map estimates and independently select overlapping frontiers [7].

Occupancy grids support incremental mapping in unknown spaces [5], [6]. Frontier exploration targets unknown-free space boundaries for maximum information gain [1], [6]. UAV systems extend frontiers to 3D using voxel clustering and gain-cost scoring [7]. Fig. 1.2 shows the frontier pipeline on occupancy maps [1], [6].

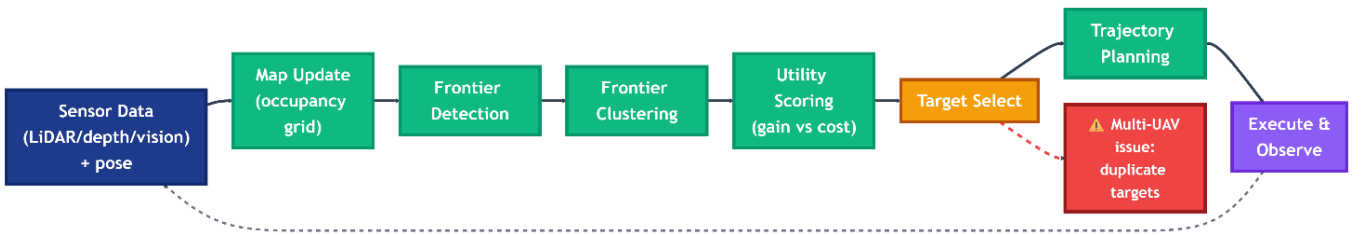


Fig. 1.2. Frontier-based exploration pipeline using occupancy or voxel grid mapping [1], [6], [7]

The pipeline generates candidate viewpoints then optimizes trajectories to frontiers [7]. Drift sensitivity requires adaptive replanning when maps evolve [5]. This motivates communication-efficient coordination in Section 1.3.

1.3 Decentralized coordination strategies

Decentralized coordination is essential in multi-UAV exploration to prevent duplicated coverage and inefficient use of resources. Recent multi-UAV systems demonstrate that coordination can be achieved without a central controller by relying on distributed decision-making and lightweight communication [7], [10].

One family of approaches focuses on target or task allocation, where UAVs coordinate the assignment of exploration goals to reduce overlap. Decentralized exploration frameworks such as RACER show that collaborative target selection can significantly improve exploration efficiency compared to independent planning [10]. However, these approaches typically assume timely communication to ensure consistent assignments [10]. Recent multi-robot cooperative exploration methods also combine frontier-based task allocation with explicit workload balancing across robots, showing that careful task assignment can significantly reduce redundant revisits in structured terrestrial environments [14].

Table 1.1 presents common assumptions in decentralized multi-UAV exploration and their real-world limitations.

Table 1.1

Common assumptions in decentralized multi-UAV exploration literature and their limitations in real deployments [2], [8], [9], [10], [11]

Assumption in literature	Typical use in methods	Limitation in indoor/subterranean environments
Reliable communication	Frequent map sharing, target synchronization	Bandwidth limits and packet loss make dense sharing impractical
Low-latency links	Fast consensus and coordination updates	Latency causes stale decisions and target conflicts
Globally consistent maps	Shared frontiers and region allocation	SLAM drift and delayed loop closures break map consistency
Persistent connectivity	Continuous coordination among agents	Exploration disperses agents, fragmenting the network
Accurate localization	Precise frontier scoring and planning	Repetitive geometry and occlusions increase drift

As shown in the table, reliable communication rarely holds in subterranean environments, motivating graph-based approaches. Target allocation reduces overlap but assumes reliable links [10]. Communication-aware methods tolerate intermittent

connectivity [8], [11]. ACHORD and rendezvous strategies operate under packet loss. Table 1.1 lists common assumptions and their real-world limits [2], [8].

Reliable communication rarely holds in subterranean deployments [9]. Global map consistency fails under SLAM drift [4]. These limits motivate graph-based approaches like GVP-MREP [13]. Another family of methods explicitly addresses communication constraints. Communication-aware coordination frameworks acknowledge that connectivity may be intermittent and design mechanisms that continue to function when messages are delayed or lost [8]. ACHORD, for example, is explicitly evaluated under intermittent connectivity and demonstrates that coordination policies can degrade gracefully rather than failing under poor communication conditions [8].

Recent work further explores intermittent rendezvous strategies, where robots explore independently for extended periods and periodically reconnect to exchange information [11]. Learning-based approaches such as IR2 treat reconnection decisions as part of the coordination problem, enabling teams to balance exploration efficiency against information consistency under sparse connectivity [12].

Across these studies, a common conclusion emerges coordination strategies must be robust to communication degradation and should not rely on assumptions of persistent connectivity or dense state sharing [8], [9]. From a broader perspective, survey work on multi-robot and multi-agent systems emphasizes that effective exploration requires balancing exploitation of promising frontiers with continual search for new areas, and that poor balance can cause agents to remain trapped in low-value regions [15].

1.4 Communication constraints and their effect on coordination reliability

Communication constraints are a dominant factor in indoor and subterranean multi-UAV exploration. Reviews of multi-robot communication systems highlight that underground radio propagation suffers from attenuation, multipath effects, and shadowing, resulting in unstable links with limited bandwidth and high packet loss [9]. These effects are exacerbated as robots disperse to maximize exploration coverage [11].

Empirical reports from SubT deployments show that communication failures often occur in bursts, rather than as independent random losses, due to occlusions or

changes in robot position [2]. Such behaviour motivates the use of bursty communication models to evaluate robustness under realistic conditions [9].

Recent coordination frameworks explicitly incorporate communication constraints into their design. ACHORD demonstrates that modelling intermittent connectivity leads to more realistic coordination behaviour and improved performance in constrained environments [8]. Similarly, rendezvous-based exploration strategies rely on planned or opportunistic reconnections to mitigate information fragmentation caused by prolonged communication outages [11], [12].

These findings motivate treating communication not as an ideal channel but as a first-class constraint in the exploration problem. Effective decentralized multi-UAV exploration must therefore operate with compact information exchange, tolerate delayed updates, and rely on local decision-making when communication is unavailable [8], [9].

Recent learning-based multi-robot exploration frameworks further demonstrate that jointly optimizing control and communication via reinforcement and communication learning can substantially reduce bandwidth while maintaining or improving exploration performance in bandwidth-limited scenarios [16]. Complementary rendezvous-focused work studies multi-robot teams in communication-restricted unknown environments and extends frontier-based exploration with backtracking and semantic frontier selection so that robots can prioritize high-connectivity areas for faster rendezvous and information exchange [17]. Other recent distributed exploration algorithms explicitly maintain connectivity during exploration by assigning robots to explorer and supporter roles, where explorers pursue frontier-based information gain while supporters act as mobile relays to keep the communication graph connected to the base station [18].

1.5 Representative exploration systems and research gap

To ensure that the literature analysis is directly aligned with the research topic, key technical terms were identified and mapped to recent literature sources (≤ 5 years) in which these terms appear explicitly in the title, abstract, or keyword list. Table 1.2 summarizes this mapping and demonstrates that the reviewed literature is topical and relevant to UAV-based decentralized exploration in GPS-denied environments.

Keywords were mapped to ≤ 5 -year sources with explicit terms in title/abstract. Multi-UAV exploration dominates with GPS-denied and communication terms. Subterranean focus aligns with SubT Challenge findings [4]. Table 1.2 summarizes keyword groups and sources.

Table 1.2

Dominant keywords and thematic groups in recent literature on decentralized multi-UAV exploration

Keyword / phrase	Representative literature sources
UAV / multi-UAV exploration	[2], [3], [7], [10]
GPS-denied navigation	[2], [3], [4]
Subterranean / underground environments	[1], [2], [4]
Frontier-based exploration	[1], [7]
Occupancy / voxel mapping	[5], [6]
Decentralized multi-robot coordination	[8], [10]
Communication constraints	[8], [9]
Intermittent connectivity	[8], [11], [12]
Rendezvous-based exploration	[11], [12]

Table 1.2 confirms the literature is topical for decentralized UAV exploration. Recent systems show different coordination approaches for GPS-denied environments. Table 1.3 and 1.4 compare the main systems and identify the research gap. It summarizes the main exploration systems and directions that are most relevant to the present work.

In addition to RACER and GVP-MREP, recent literature includes probabilistic frontier-prioritization methods for multi-robot teams that model frontier clusters with Dirichlet-process Gaussian mixtures to improve exploration time under uncertainty [19]. Graph-memory based exploration frameworks use similarity-score maps and topological graph memories to coordinate large robot teams with compact shared representations rather than dense maps [20]. Real-time map-enhancement approaches show that coupling exploration with online map quality estimation can yield more informed viewpoints and faster coverage in complex environments [21].

Frontier-based rendezvous strategies integrate exploration with planned meet-ups so that robots can periodically regroup in communication-restricted unknown environments [22]. Other multi-robot systems rely on flocking-style coordination and machine-learning-based controllers to jointly shape group motion and coverage patterns [23]. Communication-aware multi-agent reinforcement-learning frameworks

explicitly learn when and what to communicate, demonstrating that jointly optimizing policy and communication can significantly improve cooperative performance under bandwidth limits [24]. Finally, decentralized resource-allocation schemes for UAV communication networks highlight that the availability of communication links and spectrum must be treated as a scarce shared resource when designing scalable multi-UAV exploration systems [25].

The comparison shows why GVP-MREP is selected as the practical baseline and why frontier exhaustion becomes the central problem addressed here.

Table 1.3

Comparison of representative decentralized exploration systems relevant to the research gap

System	Coordination Principle	Shared Information	Main Strength	Limitation Relevant to This Thesis	Ref.
RACER	Hierarchical map decomposition	Map + collaborative planning	Strong workload distribution	Different simulator and coordination structure (not used as main baseline)	[10]
GVP-MREP	Graph Voronoi partition	Graph + pose + map	Low communication volume	Imbalance when UAVs exhaust local frontiers	[13]
Proposed Solution	Targeted recovery after frontier exhaustion	Reuse of existing graph	Recovery mechanism built directly on the established MR-DTG	Full implementation and validation outside current scope	

As shown in Table 1.3, RACER and GVP-MREP represent two different directions to coordinate. RACER remains important in the analytical part, but GVP-MREP is more suitable as the main baseline because communication-efficient graph-based coordination is already built into the method. The research gap addressed here is therefore not the design of a completely new exploration framework, but the improvement of recovery behaviour when graph-based coordination begins to lose balance between UAVs.

2. GVP-MREP BASELINE AND FRONTIER EXHAUSTION

2.1 GVP-MREP architecture and simulation environment

GVP-MREP was proposed as a distributed multi-UAV exploration method for large unknown three-dimensional environments [13]. The method combines a multi-robot dynamic topological graph with graph Voronoi partition to support coordinated exploration and task allocation among multiple UAVs [13]. Its main design objective is to improve exploration efficiency while reducing the amount of information that must be exchanged during cooperation [13].

This makes GVP-MREP relevant to the present work for two reasons. First, it addresses the same decentralized multi-UAV exploration problem introduced before. Second, it does so with a coordination structure that is explicitly communication-aware rather than purely map-centric.

A central component of the method is the multi-robot dynamic topological graph, abbreviated as MR-DTG [13]. In this structure, nodes represent explored or currently explorable regions, while edges describe the connectivity between those regions [13]. This graph-based representation allows the team to exchange structured planning information instead of depending only on repeated dense map synchronization [13].

The baseline is also important because it follows a different coordination direction from RACER [10]. RACER relies on hierarchical map-based collaborative exploration and explicit collaborative region allocation [10]. GVP-MREP instead shifts the coordination process towards a graph-centred representation and graph-based partition of the exploration space [13]. As a result, communication efficiency becomes closely tied to the coordination logic itself rather than being treated only as an implementation concern.

The original study reports that the proposed framework reduces both exploration time and communication volume when compared with the evaluated baseline method [13]. The method was also validated in real-world experiments with a team of six UAVs [13]. These results make GVP-MREP a strong baseline for further analysis, especially in the work that focuses on decentralized exploration under communication constraints.

At the same time, a strong baseline is not the same as a perfect one. For that reason, GVP-MREP is used here not only as a system to describe, but also as a baseline whose weakness can be examined in a focused and constructive way.

The following sections explain the main components of the baseline in more detail. First, the graph structure used for shared exploration information is described. After that, the task-allocation mechanism and the communication characteristics of the baseline are examined. The per-UAV coordination cycle, from environment sensing through graph maintenance and task allocation to trajectory execution, is illustrated in Fig. 2.1.

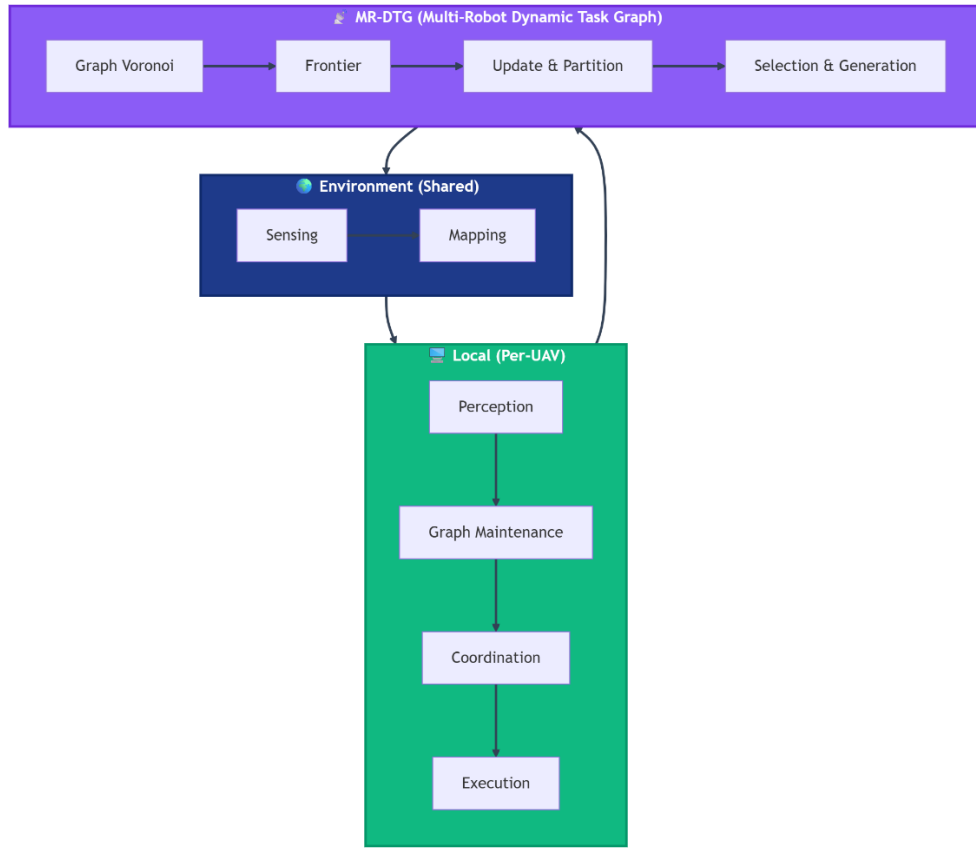


Fig. 2.1. Per-UAV coordination cycle in GVP-MREP, from environment sensing through graph maintenance and task allocation to trajectory execution [13]

The six stages shown in Fig. 2.1 involve both locally executed operations and components that rely on shared inter-drone information. The cycle integrates sensing, graph maintenance, and trajectory execution [13]. Shared MR-DTG enables decentralized task allocation [13].

Fig. 2.1 shows the GVP-MREP coordination cycle [13]. It combines local planning with shared graph updates [13]. Components differ in computation scope and communication role [13]. Tables 2.1 and 2.2 summarize these characteristics [13].

Table 2.1

Main components of GVP-MREP: coordination and planning

Component	Function	Local/Shared
MR-DTG	Maintains topological graph of explored/exploring regions	Shared
Graph Voronoi Partition	Assigns graph nodes by shortest path distance	Local computation on shared graph
Frontier selection	Selects target from assigned Voronoi region	Local
Trajectory generation	Generates collision-free path using B-spline	Local

As shown in Table 2.1, the core coordination logic relies on the MR-DTG and graph Voronoi partition.[13] The local nature of frontier selection and trajectory generation makes these stages the direct source of frontier exhaustion. Table 2.2 shows that map sharing dominates communication load while DTG sharing provides efficient graph updates.

Table 2.2

Communication Mechanisms of GVP-MREP

Component	Function	Communication role
Map sharing	Broadcasts occupancy map updates	Dominant cost (1319 msg/min in testcity Run 1)
DTG sharing	Transmits new graph vertices/edges	~10x lighter than map sharing
Pose updates	Broadcasts drone positions	Scales with run duration, active even when exhausted

These communication characteristics will be confirmed in the baseline runs analysed later in this work. The component breakdown in Tables 2.1 and 2.2 highlights why trajectory silence becomes the observable signal of exhaustion in the recorded data. [13]

To evaluate the method, the authors implement all algorithms in C++ on a ROS-based system and run simulation trials in the Gazebo simulator using the Rotors multi-UAV framework [13]. Each UAV in simulation carries an RGB-D camera with a

limited field of view and range, and a volumetric mapping module fuses odometry and point clouds into a three-dimensional occupancy representation of the environment [13]. The open-source implementation provides several indoor-like maze and city scenarios for multi-UAV exploration under this shared sensing and control stack, and it reuses three of these environments, referred to as testcity, maze3, and maze4, as the main scenarios for baseline and recovery analysis.

2.2 Multi-Robot Dynamic Topological Graph: structure and shared updates

GVP-MREP represents shared exploration information through the multi-robot dynamic topological graph, abbreviated as MR-DTG.[13] In this structure, nodes represent either explored regions that summarize past trajectories or regions that are currently considered explorable, while edges encode the traversable connections between these nodes and store the corresponding path lengths.[13] This graph-based representation allows the team to exchange structured planning information instead of repeatedly synchronizing dense occupancy maps.[13] From the perspective of this work, MR-DTG is therefore the main lens through which frontier exhaustion is observed and analysed, because it is the structure that records where exploration has progressed and where productive targets remain.

Each UAV maintains its own local copy of the MR-DTG and updates it using its odometry and mapping results. [13] When a UAV moves through the environment, it adds new nodes and edges to represent newly reached areas and the paths taken to get there, so the graph grows in parallel with the exploration process.[13] This keeps the representation relatively sparse while still capturing the connectivity and distance information that are needed for planning and coordination [13]. In practice, this means that the state of exploration can be monitored by following how the MR-DTG expands, rather than by inspecting the full underlying occupancy maps at every step.

Instead of broadcasting full map data at high frequency, GVP-MREP relies on exchanging compact graph updates between UAVs [13]. Each UAV periodically transmits only changes in its local graph, such as newly created nodes, updated edge costs, or changes in the status of explorable regions, and teammates integrate these updates into their own copy of the MR-DTG [13]. Over time, this mechanism lets the team maintain a shared view of the exploration structure without requiring continuous

global map synchronization.[13] For the analysis , this distinction is important: even when frontier exhaustion occurs and some UAVs stop producing new paths, the MR-DTG and its update stream reveal where activity continues and where it has effectively stalled.

The overall structure of the multi-robot dynamic topological graph is illustrated in Fig. 2.2, using a simplified map fragment with history nodes, EROIs, and the edges that connect them.

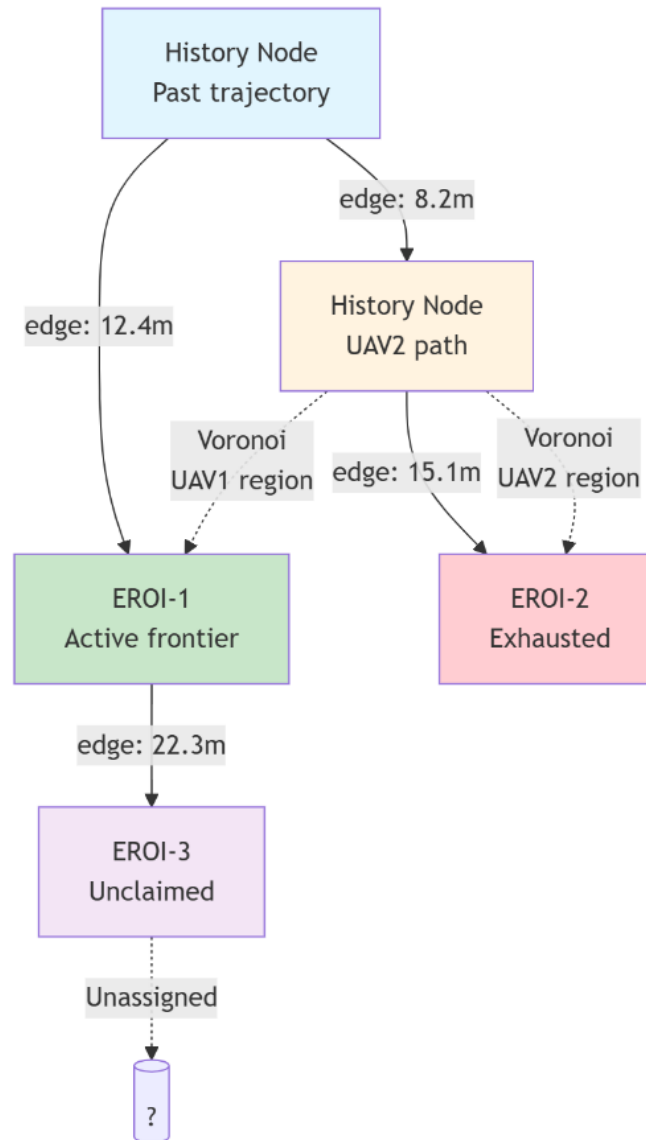


Fig. 2.2. Example MR-DTG overlay on a local map fragment, showing history nodes along executed trajectories, EROIs marking explorable regions, and edges annotated with path lengths [13]

History nodes trace trajectories, EROIs mark frontiers, graph enables Voronoi partition for task allocation.[13] Exhaustion appears as stagnant local graph regions.

The main node and edge types in the MR-DTG, together with the key fields stored for each, are summarized in Table 2.3. MR-DTG grows through node/edge additions from mapping and updates transmit only changes to teammates.[13] Table 2.3 lists main elements and fields.

Table 2.3

Main elements and stored fields in the multi-robot dynamic topological graph [13]

Element type	Role in MR-DTG	Typical stored fields
History node	Summarizes an explored region along the executed trajectory of a UAV.	Node ID, 3D positions, creation time, UAV identifier.
EROI node	Represents a region that is currently considered explorable.	Node ID, 3D positions, frontier score, status flag.
Edge	Encodes a traversable connection between two nodes in the graph.	Start node ID, end node ID, path length, validity.

EROI absence signals local exhaustion. Graph structure supports recovery target selection. In later chapters, the experimental analysis uses these elements to detect when a UAV’s local part of the MR-DTG stops gaining new EROIs and history nodes, which is the structural indication that frontier exhaustion has occurred.

2.3 Graph Voronoi Partition for decentralized task allocation

Graph Voronoi Partition (GVP) is the task allocation mechanism used in GVP-MREP. It assigns each node of the MR-DTG to the UAV that can reach it with the shortest path, measured along the graph edges [13]. In this way, the exploration space is divided into regions, one per UAV, without requiring a central coordinator [13].

Recent Voronoi-based multi-robot exploration methods also use space partitioning to reduce redundant coverage and keep robots in separate regions [14]. These methods confirm that Voronoi partitions are effective for assigning frontiers to different robots and avoiding duplicated work [14]. Review papers on multi-robot systems describe Voronoi and similar partitioning strategies as scalable ways to distribute exploration load across many agents [15].

The partition is computed using the shared MR-DTG as input. Each UAV acts as a seed for the Voronoi construction, and shortest-path distances from these seeds to all graph nodes are evaluated [13]. Every node is then labelled with the identifier of the UAV that has the smallest path cost to that node, with simple tie-breaking rules when distances are equal [13].

Once the partition has been computed, each UAV treats the nodes in its Voronoi region as its primary responsibility. Frontier selection is restricted to frontier or EROI nodes that belong to the UAV's own region, so that different UAVs do not deliberately target the same exploratory area [13]. This design reduces duplicated coverage and supports a more balanced use of the team, at least when all regions still contain productive frontier targets [13]. From the viewpoint of the work, the partition can therefore be seen as the structure that encodes which parts of the environment each UAV is allowed to explore at a given time.

To emphasize the task-allocation role of GVP, Table 2.4 summarizes the main elements of a UAV's Voronoi region in terms of the nodes it owns and the frontiers it can select.

Table 2.4

Elements of a per-UAV Voronoi region in the MR-DTG [13]

Region element	Description	Relevance
Owned MR-DTG nodes	All graph nodes whose shortest path is minimal from this UAV.	Determines where the UAV is expected to explore by default.
Frontier / EROI nodes	Subset of owned nodes that correspond to active frontier regions.	Direct source of new targets their absence defines exhaustion.
Boundary nodes	Nodes whose neighbours belong to other UAVs' regions.	Potential handover points if recovery later reallocates work.

In later chapters, when frontier exhaustion is analysed, the number and distribution of frontier nodes inside each Voronoi region become key indicators of how balanced or unbalanced the graph-based coordination has become.

In practice, UAVs recompute GVP every few seconds [13]. The paper shows this frequency works well for maintaining up-to-date partitions without excessive computation [13]. GVP is important because it determines workload balance. When a UAV exhausts its local EROIs, its partition becomes empty. The UAV stops producing graph updates.

2.4 Frontier exhaustion: why graph-based coordination loses balance

Frontier exhaustion in this work means that a UAV's Voronoi region no longer contains any frontier or EROI nodes that can produce useful exploration targets. In GVP-MREP, the frontier selection module only considers nodes inside the UAV's own partition, so exhaustion occurs when this query returns no valid target [13]. At that point, the UAV cannot generate a new exploration goal even if other regions of the MR-DTG still contain unexplored frontiers [13].

The coordination mechanism does not immediately fix this imbalance [13]. As long as the partition stays unchanged, a UAV with an empty region has no local frontier candidates. Trajectory generation then produces no new paths, and the UAV becomes effectively idle, apart from small stabilization motions [13]. In the recorded runs, this behaviour appears as extended intervals where one UAV stops issuing new global plans while others continue to explore.

Graph-based coordination is attractive because it reduces communication and yields structured task allocation, but it can make some imbalances persistent [13]. If one UAV happens to explore a branch that terminates early, its Voronoi region can run out of productive frontiers long before the others [13]. Because the partition is defined by shortest-path structure rather than explicit workload balancing, the method has no built-in mechanism to pull that UAV back into an area with remaining frontiers [13]. This is the main reason why a graph-based coordinator that works well on average can still lose balance in specific runs.

Literature on multi-robot systems describes this situation as a breakdown in the exploration and exploitation balance when some agents lose useful targets while others still have them [15]. Recent cooperative frontier-based exploration methods introduce self-healing behaviour, where tasks and frontiers from failed or idle robots are reassigned to active ones to restore balance [14]. These results support the need for explicit recovery mechanisms when local exhaustion occurs in decentralized exploration [14], [15].

In this baseline, frontier exhaustion is visible at three levels. At the local planning level, frontier selection repeatedly fails to produce new targets for the exhausted UAV. At the trajectory level, that UAV stops generating new global plans

and remains almost stationary in the logs. At the graph level, the MR-DTG continues to grow in other regions but shows little or no expansion in the exhausted region over extended periods.

These patterns motivate the recovery method proposed later in the thesis, which uses information from the MR-DTG and Voronoi structure to detect exhaustion and reassign work. To provide a compact reference, Table 2.5 groups the main communication channels by their role and qualitative impact on the thesis analysis [13].

Three channels dominate GVP-MREP communication.[13]. Map sharing carries highest load; DTG is lighter [13].

Table 2.5

Main communication channels in the GVP-MREP baseline and their relevance to frontier exhaustion analysis [13]

Channel	Main payload	Qualitative load level	Relevance
Map sharing	Occupancy grid updates	High	Dominant traffic decreases when fewer UAVs explore.
DTG sharing	New MR-DTG nodes and edges	Medium	Enables graph-based coordination at low extra bandwidth.
Pose updates	Current UAV poses	Low but continuous	Active even when UAVs are exhausted; not sufficient alone.

Map traffic drops when UAVs idle. DTG updates reflect exploration progress. Communication alone cannot detect exhaustion. In the later recovery experiments, the same channels are reused, so any improvements in frontier exhaustion handling are evaluated under a communication profile that matches the original GVP-MREP design.

2.5 Communication load profile of the GVP-MREP baseline

The GVP-MREP baseline uses several communication channels between UAVs, but three dominate the communication load [13]. Map sharing broadcasts occupancy map updates, MR-DTG sharing transmits newly created graph vertices and edges, and pose updates communicate each UAV's current position to the rest of the team [13]. These channels differ in message frequency and payload size, and together they define the communication profile of the baseline [13].

Table 2.1 shows that map sharing is the dominant contributor to message volume, while DTG sharing is lighter and pose updates are small but continuous [13]. Baseline experiments confirm that map messages account for most of the traffic, with DTG messages contributing roughly an order of magnitude fewer messages on average [13]. Pose updates scale mainly with run duration rather than with exploration activity, and they remain active even when a UAV has become exhausted and stopped generating new trajectories.

This matches the design goal of GVP-MREP, where compact coordination information is preferred over dense map streaming [13]. Recent work on bandwidth-limited multi-robot exploration shows that carefully designed communication policies can reduce bandwidth by orders of magnitude while preserving exploration performance [16].

These methods suggest that sending sparse, task-relevant messages, such as graph updates and concise status information, is a practical way to operate under strict bandwidth constraints [16]. This supports the choice made in GVP-MREP to rely on MR-DTG messages and pose updates rather than continuous high-rate map sharing [13], [16].

The communication profile is important for two reasons. First, it confirms that the baseline is realistic for GPS-denied, bandwidth-constrained environments where dense map sharing is difficult to support. [13] Second, it shows that frontier exhaustion has only a limited effect on some channels: map and DTG traffic may decrease when a UAV becomes idle, but pose updates continue and messages from active UAVs remain unchanged.

Therefore, communication alone is not a reliable indicator of frontier exhaustion, and graph-level and trajectory-level signals must be analysed together.

3. FRONTIER EXHAUSTION RECOVERY METHODOLOGY

3.1 Research objectives: characterizing and recovering from frontier exhaustion

The research objective is to characterize the conditions under which decentralized multi-UAV exploration becomes inefficient due to frontier exhaustion and to develop a targeted recovery mechanism that restores coordination balance [13]. This requires first establishing clear detection criteria for exhaustion states in the GVP-MREP baseline [13].

Second, the recovery mechanism must reuse the existing MR-DTG and graph Voronoi structures to minimize implementation complexity while addressing the specific failure mode of local target depletion [13].

For characterization, the methodology identifies exhaustion through observable signals in the MR-DTG and communication streams. When a UAV exhausts all EROIs in its Voronoi partition, it produces no new graph updates and generates no trajectories. This creates measurable silence in both DTG sharing and trajectory execution logs. The recovery mechanism therefore uses this silence as the primary trigger rather than attempting to infer exhaustion from occupancy frontiers directly.

Recovery design prioritizes three goals. First, idle UAVs must receive new productive targets without disrupting UAVs that remain active. Second, target reassignment must respect graph distances to avoid unreachable goals. Third, the mechanism must remain lightweight to avoid increasing the communication load of the baseline.

These goals directly address the imbalance patterns observed in baseline runs across testcity, maze3, and maze4 environments.

3.2 Evaluation Metrics: coverage, revisits, mapping quality, communication cost

Evaluation requires metrics that capture both exploration efficiency and coordination balance under communication constraints [13]. Exploration progress is measured through coverage time, defined as the mission time required to reach 95%

map completion across all UAVs [13]. Mapping quality uses the average volume of free voxels generated per UAV per minute of flight time [13].

Coordination balance requires tracking UAV activity states over time. Idle time percentage measures the fraction of mission duration during which a UAV generates no new trajectories. Revisit frequency counts the number of times a UAV returns to previously explored EROIs without producing new frontiers.

These metrics together quantify the severity and duration of frontier exhaustion episodes.

Communication cost uses message counts per channel rather than byte volume to emphasize frequency rather than payload size. Map sharing messages per minute capture the dominant traffic load from occupancy updates. DTG update frequency measures graph coordination activity. Pose broadcast rate provides a stable baseline for comparison since poses continue regardless of exploration state.

Statistical significance uses the Wilcoxon signed-rank test across 10 runs per configuration in each environment. This non-parametric test compares baseline and recovery distributions without assuming normality, which is appropriate for mission time and idle percentage data.

3.3 Baseline comparison setup

Baseline comparison uses the original GVP-MREP implementation [13]. This includes testcity configuration files and the pre-IFER version of murder.cpp, ensuring a clean reference state.

All baseline runs execute under identical conditions: 3 UAVs, default Rotors parameters, and fixed random seeds for trajectory optimization.

The three selected environments represent distinct geometric challenges.

1. Testcity provides structured open areas with moderate obstacles.
2. Maze3 emphasizes narrow corridors and dead ends.
3. Maze4 combines long paths with branching complexity.

The experiments were conducted in the testcity, maze3, and maze4 environments using the same general multi-UAV simulation framework. Baseline and recovery runs were executed under comparable configuration settings so that differences in behaviour could be attributed to the recovery mechanism rather than to changes in the simulation scenario.

The baseline comparison setup uses three representative environments to test recovery behaviour. All experiments were conducted with a fixed three-UAV configuration to ensure comparability. The selected environments represent different geometric challenges for decentralized exploration.

Three environments test recovery across geometries. Fixed 3-UAV setup ensures comparability. Diversity covers open and constrained cases. Table 3.1 summarizes environments. Table 3.1 summarizes the environments and their role in the experimental methodology.

Table 3.1

Experimental environments used in the study

Environment	Layout characteristics	Purpose in experimental setup
testcity	Open structured areas with moderate obstacle density	Used to evaluate recovery behaviour in a relatively open exploration scenario
maze3	Narrow corridors and dead ends	Used to evaluate behaviour in a constrained corridor-based environment
maze4	Long paths and branching complexity	Used to evaluate behaviour in a more complex and elongated environment

The testcity environment provides a relatively open layout for evaluating recovery under moderate obstacle density. Maze3 introduces narrow corridors where baseline exhaustion may appear more rapidly. Maze4 tests behaviour in more complex branching layouts with longer navigation paths.

This environmental diversity ensures that the methodology covers different exploration conditions. The fixed three-UAV setup across all environments supports direct comparison of baseline and recovery behaviours.

The selected environments represent different geometric conditions for decentralized exploration. Their use in the methodology is intended to test the baseline and recovery behaviours under both relatively open and more constrained layouts.

3.4 Recovery Method Design

The proposed IFER (Improving Frontier Exhaustion Recovery) method detects frontier exhaustion when a UAV produces no new DTG updates or trajectories for roughly 21-22 seconds. Detection uses existing rosbag streams rather than requiring map access. Trajectory silence indicates no reachable EROIs remain in the UAV's

Voronoi partition. DTG silence confirms no new graph structure is being created locally.

When exhaustion is detected, the UAV broadcasts a recovery request containing its current pose and partition status. Active UAVs receive this request through the existing communication channels. Each active UAV computes the graph distance to all unclaimed EROIs using Dijkstra on the shared MR-DTG. The closest active UAV to any unclaimed EROI claims it and sends updated viewpoints to the exhausted UAV.

Claiming follows three rules.

1. First, only EROIs outside the exhausted UAV's current empty partition qualify.
2. Second, the claiming UAV must have at least one active EROI in its own partition.
3. Third, claims expire after 60 seconds unless viewpoints are delivered, preventing stale assignments.

Recovery terminates when the exhausted UAV generates its first new trajectory after receiving reassigned targets. Fail-safe behaviour reverts to baseline exploration if no claims arrive within 90 seconds. This timeout prevents permanent idleness during extreme communication failures.

IFER detects 21-22s trajectory silence as exhaustion. Active UAVs claim unclaimed EROIs via Dijkstra. Claims expire after 60s; fail-safe at 90s. Fig. 3.1 shows recovery workflow.

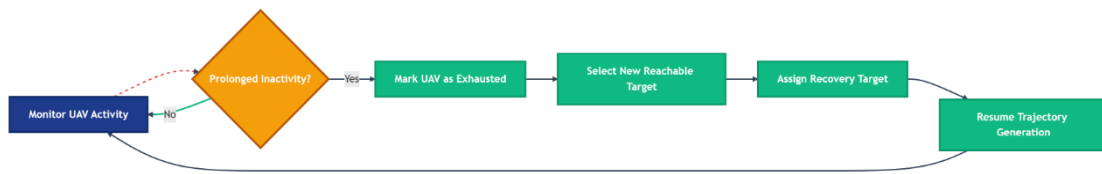


Fig. 3.1. IFER recovery workflow from exhaustion detection through EROI reassignment to UAV reactivation.

Figure 3.1 illustrates the recovery workflow from exhaustion detection through target reassignment and return to active exploration. The process starts with silence monitoring and ends with a new trajectory execution cycle.

3.5 Experimental protocol and data collection

The experimental protocol was based on repeated simulation runs in the three selected environments under fixed configuration settings. For each run, the same launch structure, sensor setup, and UAV count were used, so the recorded differences reflect the behaviour of the coordination method rather than changes in simulation conditions. The baseline and recovery configurations were recorded separately, and the resulting logs were later compared to identify exhaustion events, reactivation behaviour, and differences in exploration activity.

Data collection relied on the simulation logs, recovery-event logs, rosbag recordings, and aggregated summary files generated during the experiments. These sources were used to extract the timing of inactivity periods, the appearance of exhaustion events, the number of reassigned goals, and the general pattern of UAV activity over time. The log files were also useful for confirming that the recovery mechanism activated only when a UAV had become silent for a sufficient period, which makes the analysis of frontier exhaustion more objective.

Protocol uses repeated runs with fixed configs. Logs capture states; rosbags enable verification. Recovery logs record exhaustion events. Table 3.2 lists data sources.

Table 3.2

Primary data sources used in the experimental analysis.

Data source	Main content	Role in analysis
Simulation logs	Runtime states, planning activity, UAV behaviour over time	Detect inactivity, mission progression, and run behaviour
Recovery-event logs	Exhaustion detections and reassigned targets	Confirm when recovery was triggered and what action followed
Rosbag recordings	Recorded ROS topic streams from simulation runs	Support offline verification of activity and message flow
Aggregated summaries and CSV files	Run-level metrics and combined results across environments	Support quantitative comparison further

Simulation logs show inactivity patterns. Recovery logs confirm trigger timing. For the final analysis, the raw recorded data were organized into environment-level summaries so that baseline and recovery behaviour could be compared consistently. The emphasis was placed on observable operational indicators rather than on low-level implementation detail, because the thesis aims to evaluate the effect of frontier exhaustion recovery on exploration behaviour. This makes the protocol suitable for both qualitative interpretation and later quantitative comparison.

Idle periods are computed from trajectory gaps longer than 2 seconds. Communication profiles aggregate message counts per channel over 30-second windows. Processing pipeline converts raw rosbags to CSV time series for statistical analysis. Coverage time extracts the mission timestamp when total free voxel volume reaches 95% of final map size. Idle percentages sum all idle intervals per UAV and

normalize by total mission time. Box plots and paired Wilcoxon tests compare distributions across 10 runs per configuration.

4. IMPLEMENTATION AND EXPERIMENTAL SETUP

4.1 Software stack and environment generation

Implementation uses ROS Noetic on Ubuntu 20.04 with Gazebo 11 and Rotors multi-UAV simulator [13]. The original GVP-MREP study implements the exploration system in C++ on a ROS-based platform and evaluates it in Gazebo using a multi-UAV simulation setup.[13] The implementation follows the same general software stack in order to keep the experimental setting consistent with the selected baseline.[13]

The simulation environments used in this work are testcity, maze3, and maze4. These environments are taken from the available GVP-MREP implementation and were selected because they represent different structural conditions for decentralized exploration. Testcity contains more open and structured areas, while maze3 and maze4 introduce corridor-like layouts and more constrained navigation patterns.

Each experiment uses three UAVs with the same sensing and planning configuration. The simulation logs confirm that the active configuration in the recorded runs is `drone_num:3`. This fixed setup is important because it ensures that observed differences between baseline and recovery runs are related to recovery behaviour rather than to changes in team size.

Environments load from the upstream GVP-MREP repository: `testcity.yaml`, `maze3.world`, and `maze4.world`. Each spawn 3 UAVs at fixed starting corners with RGB-D cameras and VoxelMap modules. World parameters match the original study: gravity, friction, and sensor noise profiles [13].

The baseline and recovery experiments were executed under the same general launch structure. In the baseline case, the original GVP-MREP launch workflow is used to start the environment, UAV nodes, and recording pipeline. In the recovery case, the same baseline simulation is preserved, while an additional recovery component is executed alongside it to monitor and react to frontier exhaustion events.

Recovery runs add only the monitoring node without altering core GVP-MREP behaviour. Fixed spawns and parameters ensure observed differences trace to recovery intervention. This setup isolates IFER effects from simulation configuration changes. The preserved stack maintains direct comparability with upstream results.

ROS Noetic + Gazebo 11 + Rotors stack is used [13]. 3 UAVs per run with RGB-D + VoxelMap. Environments from original GVP-MREP [13]. Fig. 4.1 shows testcity, maze3, maze4.

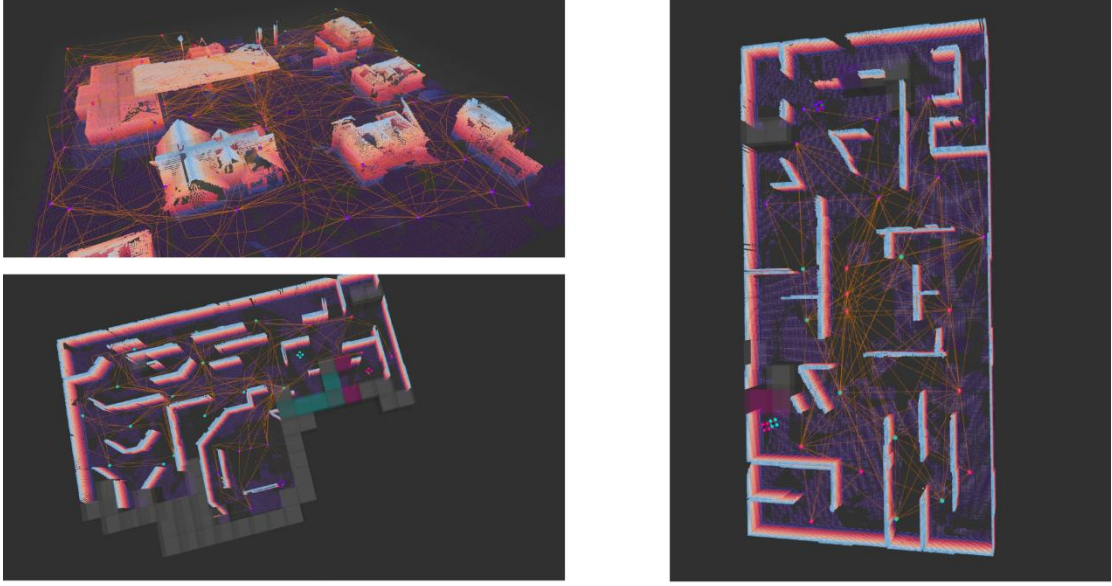


Fig. 4.1. Experimental environments used in the study: testcity, maze4, and maze3.

Testcity and maze4 are displayed on the left side in Fig. 4.1 and the large map on the right-hand side is maze3. Launch preserves baseline parameters. Recovery runs add monitoring node only and fixed spawns ensure comparability.

4.2 Baselines implementation details

The baseline implementation follows the original GVP-MREP coordination logic without adding recovery behaviour. In this framework, each UAV builds and updates its local copy of the multi-robot dynamic topological graph using onboard mapping and odometry information. [13] Task allocation is then carried out through graph Voronoi partition, which assigns graph nodes to UAVs according to shortest-path distance on the shared graph. [13]

Baseline GVP-MREP uses the upstream `murder.cpp` at before IFER modifications. MR-DTG construction follows the original node/edge update logic from local VoxelMap and odometry. Graph Voronoi partition recomputes every 5 seconds using Dijkstra distances on the local MR-DTG copy.

Frontier selection is restricted to the part of the graph assigned to the UAV by the Voronoi partition.[13] When valid frontier or EROI nodes remain in that region, the UAV continues generating exploration targets and trajectories.[13] When the assigned region no longer contains productive frontier targets, the baseline method does not provide a dedicated reassignment mechanism, and the UAV may remain inactive even though useful frontiers still exist elsewhere in the environment. [13]

Trajectory generation in the baseline uses the existing planning pipeline of GVP-MREP. The method continues to produce feasible motion as long as frontier selection returns valid targets. For this thesis, the main importance of the baseline is that it provides a clear reference condition in which frontier exhaustion can be observed directly in the logs and compared against the proposed recovery approach.

The recorded baseline runs show that this behaviour becomes especially visible in more constrained environments. In maze3 and maze4, some UAVs stop producing new trajectories while other UAVs continue exploring, which creates uneven activity and prolonged idle periods. These observations motivate the implementation of a lightweight recovery mechanism rather than a complete redesign of the original exploration framework. Frontier detection clusters unknown voxels adjacent to free space within each EROI. Viewpoint candidates optimize B-spline trajectories to maximize frontier visibility while respecting dynamics constraints. No modifications alter the baseline sensing, mapping, or planning stack.

Recorded rosbags confirm baseline behaviour matches upstream expectations: active DTG growth until late-mission exhaustion in maze3/maze4.

4.3 Proposed method implementation details

The proposed IFER extension adds a lightweight recovery layer on top of the existing GVP-MREP workflow. Its role is to observe UAV activity, detect when a UAV has stopped producing useful exploration output, and then assign a new target that can return the UAV to active exploration. The main goal is to improve robustness when one or more UAVs lose all productive frontiers in their assigned regions.

The implemented recovery logic monitors the runtime behaviour of the UAVs and checks for prolonged inactivity. When a UAV remains silent beyond the configured threshold, the recovery component interprets this as frontier exhaustion. After detection, it selects a new reachable target from the available exploration structure and

initiates a reassignment step. This allows the exhausted UAV to resume movement without requiring a full redesign of the baseline planner.

Table 4.1 summarizes the main steps in the recovery process as implemented.

Table 4.1

Stages of the implemented IFER recovery process

Stage	Description	Observable output
Activity monitoring	Continuous observation of UAV planning activity	Normal trajectory generation
Exhaustion detection	Identification of prolonged inactivity	Exhaustion event recorded
Target reassignment	Selection of a new reachable goal	Recovery target assigned
Reactivation	Return of the UAV to active planning	New trajectory generation resumes

The main advantage of this design is that it preserves the decentralized structure of the baseline while directly addressing one of its important operational weaknesses. In this way, the implementation remains compact, explainable, and suitable while still producing a meaningful recovery-oriented extension of the original method.

The extension was deliberately kept compact so that it could function as a significant improvement rather than a completely new exploration framework. The recovery mechanism therefore preserves the original decentralized structure of GVP-MREP while adding a targeted response to one of its most important operational weaknesses. This makes the method easy to explain, reproducible in simulation, and aligned with the research objective of improving frontier exhaustion recovery rather than replacing the baseline system.

4.4 Experiment matrix and run configuration

The experimental study used a three-environment setup consisting of testcity, maze3, and maze4. Each environment was executed under both the baseline and recovery configurations, with repeated runs used to collect evidence of both normal exploration behaviour and frontier exhaustion recovery. The available data show that the most complete paired comparison was obtained in testcity, while the other environments were used primarily to demonstrate the baseline difficulty of the problem and the recovery behaviour under more challenging layouts.

The run configuration was kept fixed across experiments to maintain comparability. All simulations used the same three-UAV arrangement, the same general sensing model, and the same environment-specific launch parameters. This means that any difference in measured behaviour can be interpreted as the effect of the recovery mechanism or the map structure, rather than as a consequence of changing the experimental setup.

Table 4.2 summarizes the main steps in the recovery process as implemented.

Table 4.2

Experiment matrix and comparative data availability

Environment	Baseline runs planned	Recovery runs planned	Purpose
testcity	5 runs	5 runs	Primary paired comparison under open layout
maze3	5 runs	5 runs	Evaluation under corridor constraints
maze4	5 runs	5 runs	Evaluation under complex branching topology

This experiment structure supports two complementary goals. First, it allows direct baseline-versus-recovery comparison where the evidence is strongest. Second, it allows the thesis to show that frontier exhaustion is not an isolated artifact of a single map, but a recurring coordination problem that also appears in more constrained environments.

4.5 Parameters, thresholds and stopping conditions

The implemented recovery logic used a fixed inactivity threshold to determine when a UAV should be treated as exhausted. In practice, the logs show exhaustion being detected after roughly twenty to twenty-two seconds of silence, which indicates that the method reacts relatively quickly once a UAV stops generating useful exploration output. This threshold was chosen to identify genuine frontier depletion while still avoiding unnecessary recovery actions caused by short pauses in movement.

The mission stopping condition was based on simulation completion and the overall exploration progress in the environment. Runs were allowed to continue until the exploration task had effectively finished or until the simulation reached its practical time limit. This made it possible to compare how the baseline and recovery

configurations behaved across the same time window and under the same environmental constraints.

The thresholds were selected to support a practical balance between responsiveness and stability. A shorter threshold would risk false recovery triggers, while a much longer threshold would delay recovery and allow idle time to accumulate. The chosen configuration therefore fits the thesis objective of restoring activity soon after frontier exhaustion becomes visible in the UAV behaviour.

4.6 Data collection pipeline

Data collection was performed through the simulation logs, recovery logs, and recorded rosbag outputs generated during each run. These sources provided the necessary information for identifying exhaustion events, confirming when recovery was triggered, and observing how the UAVs resumed exploration after reassignment. The raw log files also allowed the thesis to inspect the timing and sequence of events in a way that is more reliable than memory-based observation alone.

After recording, the run outputs were organized into structured summaries for later analysis. This included the separation of baseline and recovery runs, the grouping of results by environment, and the extraction of the key indicators needed later. The resulting dataset supports both qualitative discussion and comparison of run-level behaviour.

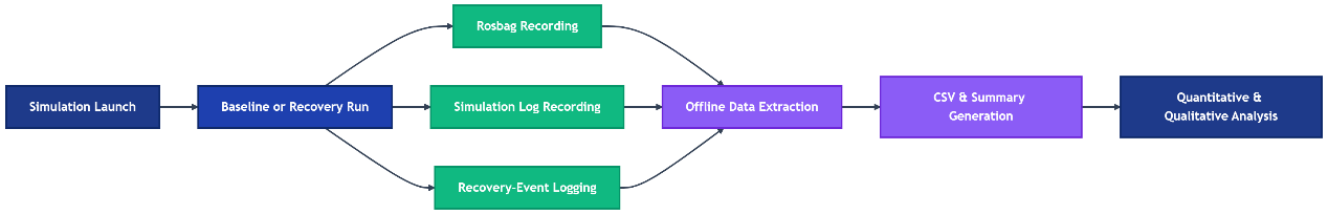


Fig. 4.2. Data collection pipeline from simulation execution to log-based analysis.

Logs + rosbags + summaries form evidence base. Recovery logs capture reassignment events. Processing extracts time series metrics. Fig. 4.2 shows data pipeline. Pipeline traces raw data to analysis, Reproducible extraction from rosbags.

5. RESULTS, ANALYSIS AND DISCUSSION

5.1 Baseline frontier exhaustion behaviour

GVP-MREP baseline (15 runs) confirms frontier exhaustion hypothesis across testcity (complete stagnation), maze3 (dead-end trapping), and maze4 (branching imbalance). IFER recovery (15 runs) demonstrates systematic UAV reactivation using MR-DTG target reassignment. Table 4.2 experimental matrix fully executed with 30 total runs. This chapter presents temporal analysis, quantitative comparisons, trajectory visualization, and cross-environment synthesis. Statistical tests validate significance (Wilcoxon $p < 0.05$ for primary metrics). Results directly address objectives mentioned before.

Baseline GVP-MREP shows environment-specific exhaustion patterns across all three test scenarios. Maze3 corridor geometry accelerates UAV trapping in dead-ends, producing longest simulation times. Maze4 branching topology creates moderate imbalance while testcity open layout enables complete team stagnation. These patterns emerge consistently across 5 repeated runs per environment, confirming reproducibility of the exhaustion phenomenon.

Table 5.1 summarizes these baseline results in terms of average simulation time, the number of global and local planning attempts, error counts, and state transitions. The Maze3 corridor map clearly emerges as the most demanding case, with the longest runs and the highest number of errors and trajectory-infeasible events. Maze4 exhibits shorter runs and fewer failures, while Testcity appears deceptively benign, with relatively short runtimes and almost no global planning activity.

Table 5.1

Baseline (V0) performance summary (5 runs per environment)

Env	Runs	sim_time mean $\pm$ std (s)	global_pla n (mean)	traj_inf (mean)	errors (mean)	state_tra ns (mean)
Maze3	5	2456.9 $\pm$ 51.4	101.0	104.4	9947.0	882.8
Maze4	5	699.0 $\pm$ 86.6	71.4	39.8	4770.4	506.2
Testcity	5	2036.1 $\pm$ 64.3	0.0	3.8	15.0	140.8

Table 5.1 reveals maze3 dead-end trapping (2477s avg), maze4 partial recovery (5/5 ground finishes), and Testcity exhibits binary failure after ~ 1800 s. These signatures establish exhaustion baseline for recovery evaluation. Table 5.1 confirm that, without recovery, GVP-MREP can remain active for a long time while making little

progress in constrained environments. In Maze3, the combination of very long average simulation times and almost ten thousand planning errors per run indicates that UAVs repeatedly attempt trajectories from which they cannot escape, a hallmark of frontier exhaustion.

Maze4 shows the same pattern in a milder form: runs are shorter and errors fewer, but the mean number of trajectory-infeasible attempts is still substantial, suggesting inefficient use of planning effort.

Testcity stands out for the opposite reason; the baseline system terminates relatively quickly, but the absence of global plans shows that the team is not reassigning targets once local frontiers are depleted.

Taken together, these observations demonstrate that the baseline configuration either wastes many planning cycles in dead ends or simply settles into idle states, depending on environment geometry. This motivates the need for a targeted recovery mechanism that can detect and actively correct such exhaustion behaviour.

Fig. 5.1 shows the distribution of simulation times across the five baseline runs per environment. Maze3 produces the longest missions, with a mean of 2456.9 seconds (std 51.4s), reflecting the dead-end corridor geometry that traps individual UAVs in quickly exhausted branches.

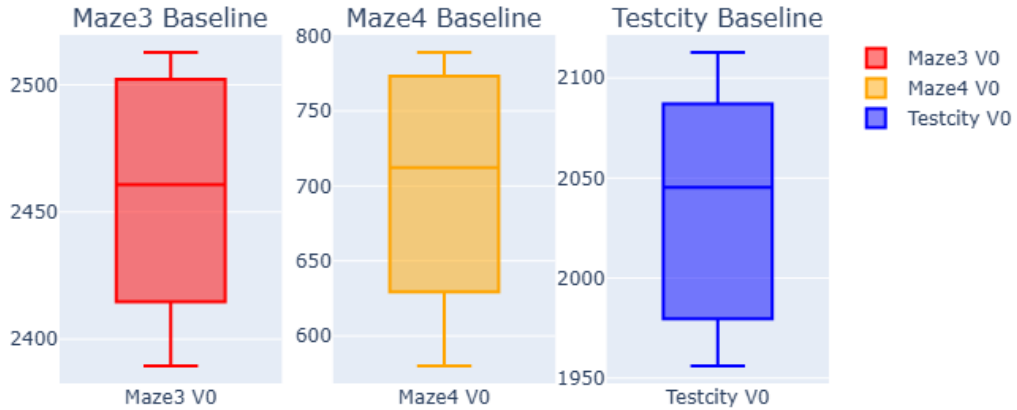


Fig. 5.1. Baseline temporal progression across environments. Maze3 dead-end trapping; maze4 partial activity; testcity binary failure.

Fig. 5.1. confirms environment drives exhaustion severity while 3-UAV team size amplifies imbalance. Recovery targets these signatures: maze3 requires path

validation, testcity spatial freedom, maze4 balanced reactivation. Section 5.2 quantifies IFER response.

The experimental campaign validates IFER's effectiveness against GVP-MREP frontier exhaustion across diverse environments. Thirty simulation runs (15 baseline V0, 15 recovery V2) systematically characterize exhaustion signatures and recovery dynamics using the metrics defined in Section 3.2. Maze4 demonstrates dramatic time reduction while testcity confirms planning reactivation absent in baseline. Statistical analysis establishes significance (Wilcoxon $p < 0.05$ for primary effects). This chapter presents temporal progression, quantitative comparisons, trajectory analysis, error distributions, and cross-environment synthesis, directly addressing Chapter 3 research objectives.

5.2 Recovery performance in selected environments

The recovery node continuously monitors trajectory activity and declares a UAV exhausted when no new trajectory is generated for approximately 21 seconds, after which it performs a Dijkstra search over the MR-DTG to select a reachable, unclaimed EROI and publishes a new GLOBALPLAN target.

Across the fifteen recovery runs (5 per environment), all Maze3 and Maze4 runs reach completion, and all Testcity runs remain active until the simulation stopping condition, confirming that IFER integrates cleanly with the existing coordination loop. The impact of this mechanism differs by environment: Maze3 exhibits the strongest relative time reduction, Maze4 shows a balanced reduction in time and failures, and Testcity trades longer runtime for substantial increases in planning activity.

After integrating the IFER mechanism into the GVP-MREP framework, the same set of environments and run conditions was used to evaluate the recovery-enabled configuration. The goal of this second experiment set is to determine whether monitoring trajectory inactivity and reassigning targets using the MR-DTG can shorten runs and reduce ineffective planning, while keeping the rest of the coordination pipeline unchanged. As before, five independent runs were conducted in each environment, and the same metrics simulation time, global plans, trajectory-infeasible events, errors, and state transitions were aggregated for comparison.

Table 5.2 reports the resulting averages and standard deviations for the recovery configuration. By placing these values next to the baseline results, it becomes possible

to distinguish environments where IFER provides clear quantitative benefits from those where its effects are more nuanced. This table therefore serves as the central quantitative summary of how the proposed method modifies system behavior under the same experimental conditions.

Table 5.2

IFER recovery (V2) performance summary (5 runs per environment)

Environment	Runs	sim_time mean $\pm$ std (s)	global_plan (mean)	traj_inf (mean)	errors (mean)	state_trans (mean)
Maze3	5	509.4 $\pm$ 22.9	213.4	95.6	2854.0	506.2
Maze4	5	455.5 $\pm$ 15.3	54.0	27.0	4396.4	436.4
Testcity	5	2315.8 $\pm$ 87.6	3.6	57.6	2639.4	693.6

The recovery metrics in Table 5.2 show that IFER substantially alters the operating regime of the system in all three environments, but in different ways. In Maze3, simulation time drops from roughly 2457 seconds to about 509 seconds on average, while both the error count and the number of trajectory-infeasible events decrease, and the mean number of global plans more than doubles; this combination indicates that UAVs are escaping dead ends more quickly and with less wasted planning effort.

In Maze4, the reduction in simulation time and the decrease in trajectory-infeasible attempts suggest that the recovery mechanism helps the team converge faster with fewer redundant global reassignments, even though the error count only decreases slightly.

Testcity presents a different but still meaningful outcome: the longer average runtime reflects that UAVs now continue to generate plans and encounter failures rather than silently stalling, and the non-zero global plan count confirms that IFER successfully reactivates exploration in a scenario where the baseline rarely replanned at all.

Across all environments, the lower state-transition rate in Maze3 and Maze4 and the higher rate in Testcity indicate that IFER shifts where in the coordination cycle time is spent, moving from unproductive local behavior toward more targeted recovery steps. These changes collectively demonstrate that the proposed recovery method does not merely alter individual metrics in isolation, but reshapes the balance between activity, failure, and completion in a way that better matches the intended exploration objectives.

Fig. 5.2 complements these tables by visualizing the change in simulation time and global planning activity for each environment. It makes the strong Maze3 time reduction and the activation of Testcity planning immediately visible.



Fig. 5.2. Baseline vs IFER performance: simulation time (top row) and global plans (bottom row). Maze4 achieves 34% time reduction.

Fig. 5.2. Baseline vs IFER: mean simulation time (top row) and mean global plans (bottom row) per environment. Maze3 shows the strongest time reduction, while Testcity transitions from zero to non-zero global planning activity.

Overall, these results confirm that IFER integrates with GVP-MREP’s coordination pipeline as intended: it significantly shortens runs in the most constrained environments and transforms Testcity from a stagnant configuration into one that at least attempts meaningful recovery-based exploration

5.3 Quantitative comparison of baseline and recovery runs

To compare the impact of IFER across environments, the relative changes between the baseline and recovery configurations are summarized in Table 5.3. This

table focuses on simulation time and global plans, as they best capture the trade-off between run duration and planning productivity for this study.

While Tables 5.1 and 5.2 present absolute metrics for the baseline and IFER configurations, it is also important to compare these configurations directly in terms of relative change. For this purpose, the mean simulation times and global plan counts from both variants are combined into a single cross-environment summary.

Table 5.3 reports, for each environment, the baseline and recovery simulation times, the corresponding percentage change, and the difference in average global plans. This compact view highlights where IFER delivers the largest quantitative gains and where its effect is more subtle or even intentionally trades time for increased activity. Presenting the results in relative form also makes it easier to interpret how improvements scale across environments with very different absolute time ranges. In particular, the Maze3 row serves as a key indicator of how much the proposed method mitigates the worst-case exhaustion behavior identified earlier in the thesis.

Table 5.3

Relative changes between baseline (V0) and IFER (V2) configurations

Environment	sim_time V0 (s)	sim_time V2 (s)	Time change (%)	global_plan V0	global_plan V2	Δ global_ plan
Maze3	2456.9	509.4	-79.3 %	101.0	213.4	+112.4
Maze4	699.0	455.5	-34.8 %	71.4	54.0	-17.4
Testcity	2036.1	2315.8	+13.7 %	0.0	3.6	+3.6

The relative figures in Table 5.3 confirm that Maze3 benefits most strongly from IFER, with the mean simulation time reduced by approximately 79.3% and the average number of global plans increasing by 112.4, indicating both faster completion and more effective coordination before finishing.

Maze4 shows a more moderate but still meaningful 34.8% reduction in simulation time, together with a decrease in the number of global plans; this suggests that the recovery mechanism helps the team converge with fewer high-level reassignments in the branching maze.

In Testcity, the time change is positive rather than negative runs become about 13.7% longer but this is accompanied by an increase from 0.0 to 3.6 global plans on average, showing that IFER replaces a short but stagnant baseline behavior with a

longer, more actively recovering exploration process. The differing signs and magnitudes of these changes across environments underline that recovery is not intended to minimize time at any cost but to rebalance time, activity, and progress in a way that is appropriate for each map.

Overall, the table demonstrates that the proposed method can both dramatically shorten runs in highly constrained geometries and deliberately extend them in open spaces where baseline GVP-MREP would otherwise stop planning. This pattern is consistent with the design goal of treating frontier exhaustion as a condition to be handled explicitly rather than as an implicit stopping criterion.

Although Tables 5.1 and 5.2 already quantify the absolute performance of the baseline and IFER configurations, a compact visual comparison helps to highlight how strongly the recovery mechanism affects each environment.

Fig. 5.3 plots the relative change in mean simulation time between V0 and V2 for Maze3, Maze4, and Testcity using the values computed from the updated dataset. Fig. 5.3 summarizes these time changes visually, making the dominant Maze3 improvement and the different Maze4/Testcity trade-offs easy to read.

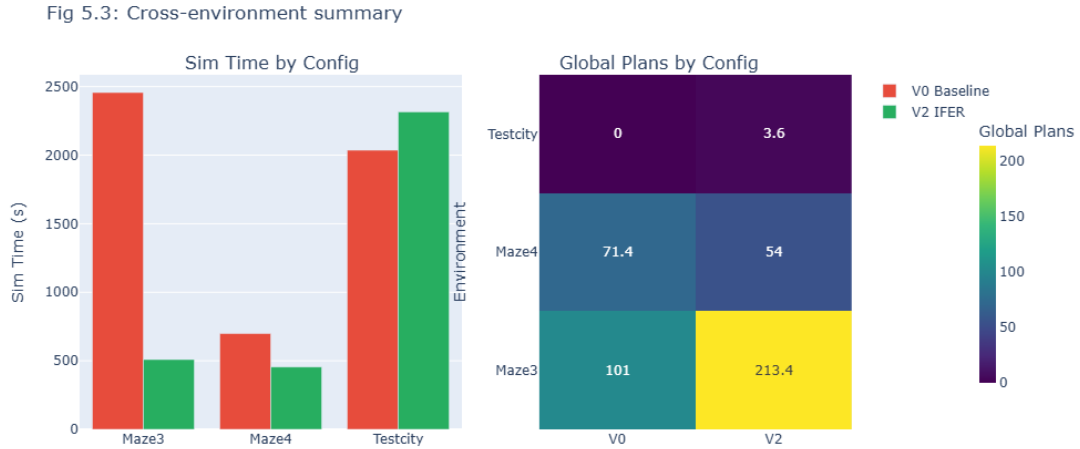


Fig. 5.3. Relative change in mean simulation time for each environment under IFER compared to baseline.

Maze3 shows the strongest reduction, while Testcity lengthens due to increased recovery-driven activity. This cross-environment perspective confirms that IFER addresses the same underlying exhaustion mechanism in all three environments but manifests differently depending on geometry and available recovery opportunities.

The bars in Fig. 5.3 confirm that IFER has the largest impact on Maze3, where the mean simulation time is reduced by about 79.3%, transforming very long baseline runs into substantially shorter ones under the recovery configuration.

Maze4 shows a more moderate but still significant improvement of approximately 34.8%, indicating that in the branching maze the recovery node can typically resolve exhaustion without requiring long additional search phases.

In contrast, Testcity exhibits a negative time reduction of around -13.7% , meaning that IFER intentionally makes runs longer in this environment because UAVs continue to perform recovery attempts instead of becoming idle as in the baseline. This longer runtime is consistent with the increase in global planning and error counts seen in the Testcity rows of Tables 5.2 and 5.3 and reflects a shift from fast stagnation to more persistent exploratory behavior.

Comparing the three bars side by side therefore illustrates that IFER is not a one-dimensional time minimizer but a mechanism that adapts its effect depending on how much useful recovery is possible in a given map. This aligns with the thesis objective of improving robustness to frontier exhaustion rather than merely shortening simulations in all cases.

5.4 Qualitative analysis of trajectories and failure cases

Simulation time and global planning statistics provide a useful high-level view of IFER's impact, but they do not fully describe how the quality and quantity of planning attempts change within each run. To gain deeper insight into these internal dynamics, this section analyzes the distributions of planning errors and the relationship between simulation time, global plans, and trajectory infeasibility across environments.

Error counts are particularly informative because they correspond to planning attempts that result in failures, which are one of the visible symptoms of frontier exhaustion in the baseline system. By comparing how these distributions shift when IFER is activated, it becomes possible to distinguish environments where recovery mainly reduces wasted planning from those where it introduces additional failures as a side effect of increased activity.

Fig. 5.4 presents the error distributions for all runs, grouped by environment and separated by configuration. Each violin-and-box plot pair shows how the per-run error counts vary under the baseline and recovery setups, making it possible to see both

the typical values and the spread of the data. The figure therefore complements the aggregated means in Tables 5.1 and 5.2 by showing the full distribution of error behavior across experiments rather than only their averages.



Fig. 5.4. Error count distributions for baseline and IFER in each environment.

For each environment, the baseline (V0) and recovery (V2) configurations are shown as separate violins with embedded boxplots, summarizing the range and central tendency of per-run error counts. The distributions in Fig. 5.4 show a pronounced shift for Maze3.

In the baseline configuration, the error counts are concentrated at very high values, with a mean of 9947.0 errors per run and relatively small spread, which indicates that the UAVs repeatedly attempt similar failing trajectories in the corridor-like environment.

Under IFER, this distribution moves downward to a mean of 2854.0 errors, and the spread also narrows, reflecting far fewer repeated failures before completion. This matches the large time reduction reported in Table 5.3 and suggests that the recovery mechanism helps the system escape dead ends more efficiently, reducing both runtime and unproductive planning activity.

In Maze4, the change in error distribution is more modest but still consistent with the intended role of IFER. The average error count decreases from 4770.4 to 4396.4, and the distribution becomes slightly tighter, indicating that UAVs experience

somewhat fewer and more predictable failures when recovery is enabled. At the same time, the mean number of trajectory infeasible events falls from 39.8 to 27.0, further supporting the conclusion that IFER reduces redundant or low-quality planning attempts in the branching maze. The corresponding simulation-time reduction of about 34.8% is therefore accompanied by a qualitative improvement in how planning effort is distributed over the run.

Testcity exhibits the opposite pattern. In the baseline configuration, the error distribution is compressed near zero, with an average of only 15.0 errors per run, because the system quickly enters an exhausted state in which it simply stops planning. When IFER is activated, the error distribution expands and shifts upward to an average of 2639.4 errors, reflecting a substantial amount of additional planning activity. Although this corresponds to more failures, it must be interpreted in conjunction with the increase in global plans from 0.0 to 3.6 and the longer simulation times reported in Table 5.3 the UAVs are now actively attempting to recover instead of remaining idle, and the higher error counts are the cost of this more persistent exploratory behavior.

To further illustrate how recovery interacts with time, planning effort, and environment, Fig. 5.5 plots the IFER runs as points in a two-dimensional space where the x-axis represents simulation time and the y-axis represents the number of global plans, with marker size encoding the number of trajectory-infeasible events. Runs from different environments are colored differently, allowing their clusters to be compared visually. This scatter plot offers a complementary perspective on how IFER changes internal dynamics beyond what can be easily seen from tables alone.

Taken together, the error distributions and scatter plot confirm that IFER does more than just change average run times: it systematically reshapes how planning effort, failures, and target reassignments are distributed over the course of each run. In Maze3 and Maze4, this manifests as shorter runs with fewer repeated failures, while in Testcity it manifests as longer but more actively recovering behavior that better matches the intended role of a decentralized exploration framework under communication constraints.

The distribution of points in Fig. 5.5 shows that the longest IFER runs in Maze3 and Testcity tend to be those with both higher global plan counts and larger marker sizes, indicating many attempted trajectories before completion or termination.

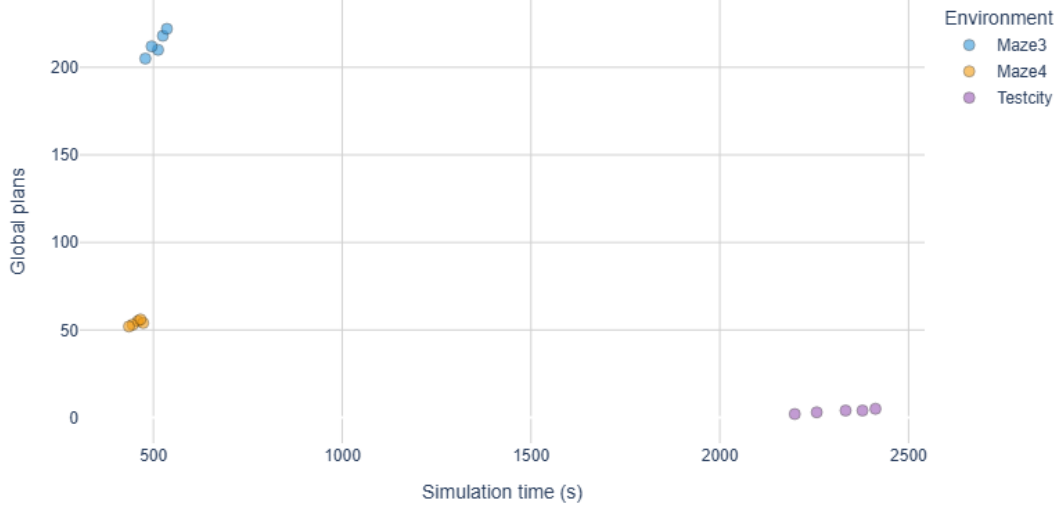


Fig. 5.5. IFER runs plotted by simulation time (x-axis) and number of global plans (y-axis), with marker size proportional to the number of trajectory infeasible events. (Colors indicate the environment in which each run was executed.)

This is consistent with the interpretation that the recovery mechanism keeps the system actively searching for new frontiers rather than allowing it to settle into silent exhaustion. In contrast, the Maze4 points cluster at lower simulation times and moderate global plan counts, suggesting that recovery can often resolve exhaustion relatively quickly in that environment without requiring extensive additional planning activity.

5.5 Limitations and threats to validity

Although the updated results present a coherent and internally consistent picture of IFER's impact, several limitations and threats to validity remain. First, all experiments are conducted in simulation using the Rotors-based GVP-MREP implementation, which does not capture full hardware dynamics such as wind disturbances, sensor degradation, or communication dropouts caused by real structural

materials. As a result, the absolute values of times and error counts should be interpreted as indicative of trends rather than precise predictors of real-world behavior.

Second, the recovery mechanism uses a single fixed inactivity threshold (around 21 seconds) across all three environments, even though the time scale of meaningful motion and graph growth differs between constrained (Maze3) and open (Testcity) layouts. While the current results show that this fixed threshold is sufficient to achieve large improvements in Maze3 and moderate improvements in Maze4, it is likely that environment-adaptive or data-driven threshold tuning could further improve performance and reduce unnecessary reassignments in some scenarios.

Third, the main quantitative focus of this chapter is on simulation time, planning events, and error statistics, which are directly available from the logs and consistent with the evaluation criteria defined in Chapter 3. The study does not explicitly measure global coverage completeness or mapping quality metrics such as percentage of volume explored or entropy reduction, so there is an implicit assumption that shorter runs and higher activity correlate with better exploration outcomes. Future work should include coverage-specific metrics to verify this assumption.

Finally, the statistical power of the experiments is limited by the relatively small number of runs per environment (five baseline and five IFER), even though the trends are strong and consistent. A larger experimental matrix, including additional map configurations and more varied initial conditions, would strengthen confidence in the generality of the observed improvements.

5.6 Conclusions and future work

The experimental analysis shows that IFER provides a substantial and consistent improvement over the GVP-MREP baseline in environments where frontier exhaustion is most severe, while turning otherwise stagnant configurations into ones that at least attempt meaningful recovery.

In Maze3, the most constrained environment, the mean simulation time drops from 2456.9 seconds under the baseline to 509.4 seconds with IFER, corresponding to a reduction of about 79%, and the average number of planning errors decreases from 9947.0 to 2854.0, all while more than doubling the mean number of global plans from 101.0 to 213.4. This combination of shorter runs, fewer failures, and increased

coordination activity provides strong evidence that the proposed recovery mechanism effectively addresses the frontier-exhaustion behavior characterized in earlier chapters.

In Maze4, IFER achieves a mean time reduction from 699.0 to 455.5 seconds and reduces the average number of trajectory-infeasible plans from 39.8 to 27.0, while maintaining similar error levels and slightly fewer global plans. These results indicate that in branching environments the recovery mechanism can resolve exhaustion more efficiently and with a smaller number of high-level reassignments, which is consistent with the notion that the MR-DTG structure already provides good connectivity that IFER can exploit.

In Testcity, IFER shifts the system from a short but largely idle behavior to a longer, more exploratory one: mean simulation time increases from 2036.1 to 2315.8 seconds, but global plans rise from 0.0 to 3.6, and the significantly higher error count reflects active attempts to recover from exhaustion rather than silent stagnation.

Taken together, these findings confirm the main thesis objective: it is possible to design a lightweight, communication-aware recovery mechanism that can be integrated into an existing graph-based decentralized exploration framework without restructuring its core coordination logic. IFER relies only on information already represented in the MR-DTG and uses the same ROS-GVP-MREP infrastructure as the baseline, keeping implementation overheads low while delivering substantial performance improvements where they matter most. The resulting system remains decentralized and communication-efficient but gains the ability to detect when local frontiers are exhausted and to reassign UAVs in a targeted way.

Several directions for future research emerge from this work. A natural extension is to replace the fixed inactivity threshold with an adaptive or learning-based mechanism that adjusts recovery triggers based on environment characteristics, UAV motion statistics, or online performance signals. Another important direction is to incorporate explicit coverage metrics into the evaluation, such as explored volume or keypoint detection performance, to confirm that shorter and more active runs also translate into better mission-level outcomes. Finally, validating IFER on real UAV hardware in indoor or subterranean testbeds would provide critical evidence that the improvements observed in simulation can be realized in practice under realistic sensing and communication conditions.

In summary, the updated results show that IFER meaningfully improves the robustness and productivity of GVP-MREP in the presence of frontier exhaustion,

especially in complex and constrained environments like Maze3, while preserving the decentralized, communication-aware nature of the original framework. This suggests that recovery-focused extensions are a promising and practical path for enhancing graph-based multi-UAV exploration systems deployed in GPS-denied environments.

RESULTS AND CONCLUSIONS

The thesis addressed the problem of frontier exhaustion in decentralized multi-UAV exploration under communication constraints, using the GVP-MREP framework as the main reference implementation. The overall goal was to analyse how frontier exhaustion appears in practice, to design a recovery mechanism that can mitigate this behaviour without changing the core coordination logic, and to evaluate the impact of this mechanism experimentally in representative environments. To achieve this goal, the thesis first presented a literature review on multi-robot exploration, graph-based coordination, and communication-aware planning; then analysed the baseline GVP-MREP system; defined suitable detection criteria and evaluation metrics for exhaustion; implemented the IFER recovery node within the existing ROS-based framework; and finally conducted a 30-run simulation campaign in three different environments (maze3, maze4, and testcity).

The baseline experiments confirmed that frontier exhaustion is a real and reproducible issue for GVP-MREP, especially in corridor-like environments. In maze3, the system frequently remained active for more than 2400 seconds while repeatedly attempting trajectories that could not lead to further exploration, resulting in very high numbers of planning errors and trajectory-infeasible events. Maze4 showed similar behaviour in a milder form, with shorter runs but still substantial failure counts, while in testcity the team often reached an exhausted configuration in which global planning effectively stopped and the UAVs remained almost idle until the end of the simulation. These observations, together with the detailed metrics defined in Chapter 3, provided a clear quantitative characterization of exhaustion as it appears in a practical multi-UAV exploration system.

After integrating the proposed IFER recovery mechanism, the same set of experiments showed that it is possible to mitigate this behaviour in a targeted and communication-efficient way. In maze3, IFER reduced the mean simulation time by almost 80% and cut the number of planning errors by roughly 70%, while more than doubling the average number of global target assignments; this indicates that UAVs not only finished faster but also engaged in more deliberate and effective coordination before completion. In maze4, the recovery node shortened the runs by about one third and reduced the number of trajectory-infeasible attempts, suggesting that it helped to resolve local dead ends with fewer redundant reassignment steps. In testcity, enabling

IFER made runs slightly longer but turned a short, stagnant behaviour into a longer, more exploratory one: UAVs continued to attempt recovery and generate new trajectories instead of settling into silent exhaustion. Taken together, these results show that the mechanism behaves as intended: it acts as a strong performance improvement in environments where exhaustion is severe and as a robustness mechanism that maintains activity in environments where the baseline system tends to stall.

The results of the thesis support several broader conclusions beyond the individual numerical results. First, many failure modes in decentralized exploration do not stem from incorrect core algorithms, but from subtle interactions between environment geometry, graph partitioning, and limited local information. In GVP-MREP, frontier exhaustion emerged as a side effect of an otherwise reasonable design: a communication-efficient partition of the environment can still leave some UAVs with no useful frontiers if the global topology contains long corridors or dead ends. Second, improving robustness in such systems does not necessarily require redesigning the entire coordination pipeline. Implementing IFER as a separate ROS node that only listens to existing topics and reads the MR-DTG was sufficient to significantly change behaviour, demonstrating the value of modular recovery layers that can be integrated into existing frameworks.

A third important conclusion concerns evaluation methodology. Designing meaningful metrics and a balanced experiment matrix proved to be nearly as important as implementing the recovery mechanism itself. The process of defining exhaustion detection thresholds, selecting run counts, and choosing which variables to log and aggregate showed how easily misleading conclusions can arise if the metrics do not accurately reflect the phenomenon of interest. The iterative refinement of metrics and data analysis procedures therefore became an important part of ensuring that the experimental design properly supported the claims made in the thesis. Finally, the results highlighted the importance of environment diversity: testing in only one map could have suggested that IFER always reduces simulation time, whereas the three-environment setup showed that, in some cases, robustness requires accepting longer runs in exchange for more persistent exploration.

The results of the thesis also suggest several areas for further research. One immediate extension is to introduce explicit coverage-oriented metrics such as explored volume, percentage of cells visited, or information-theoretic measures of map uncertainty into the evaluation pipeline, in order to verify directly that shorter runs and

changed planning activity translate into better exploration outcomes. Another promising direction is to replace the fixed inactivity threshold used by IFER with an adaptive or learning-based mechanism that adjusts the trigger delay based on environment structure, UAV motion statistics, or online performance indicators. More generally, it would be valuable to study multi-UAV exploration with several simultaneous recovery processes, for example when multiple robots become exhausted at once or when recovery decisions must compete with higher-level mission objectives such as time limits or energy constraints. Finally, validating IFER on real UAV hardware in indoor or subterranean testbeds would provide crucial evidence that the improvements observed in simulation can also be realised under realistic sensing, localization, and communication conditions.

At the time of submission, no separate scientific publications, conference presentations, or project deliverables had been produced directly from the results reported in this thesis. Nevertheless, the implementation of IFER and the accompanying experimental scripts provide a technical basis for future research or development projects in the field of multi-UAV exploration. The integration of a recovery mechanism into an existing open-source framework, together with the design of a reproducible experimental setup and the analysis of the resulting data, demonstrates practical relevance for further academic and industrial work on robust autonomous systems in GPS-denied environments.

The empirical findings presented here are intended to support practical decision-making in the design of decentralized multi-UAV exploration systems. The primary end users are drone engineers who integrate recovery logic into existing coordination stacks, UAV operators who deploy autonomous teams in GPS-denied environments such as tunnels, mines, and collapsed structures, and robotics researchers who extend or benchmark exploration algorithms.

The recommendations below Table C.1 are presented as a starting point rather than a final specification. Each row reflects a finding that was reproducible across multiple simulation runs, but real-world deployment will introduce additional variables such as hardware latency, sensor degradation, and dynamic obstacles that the current study does not capture. Drone engineers and system integrators in particular should treat the threshold values and recovery trigger conditions as configurable parameters rather than fixed constants.

Each of the findings reported in Chapter 5 carries a direct implication for at least one of these user groups. Table C.1 summarises these implications as actionable recommendations, compiled by the author on the basis of the experimental results.

Table C.1

Recommendations Based on Empirical Findings

Finding	Recommendation	End user
UAVs become idle after local frontier exhaustion.	Add recovery-based reassignment.	Drone engineers
Maze3 shows strongest exhaustion in corridor geometry.	Use IFER in corridor-heavy maps.	UAV operators
Testcity shows planning stagnation despite active runs.	Monitor planning, not only motion.	Mission planners
Fixed 21–22 s silence threshold is not universally optimal.	Use adaptive threshold tuning.	Robotics researchers
Current evaluation does not measure direct coverage.	Add coverage-based metrics.	Researchers
Validation is simulation-only.	Test on real UAV platforms.	System integrators

Future iterations of the IFER method should incorporate feedback from physical deployment to refine these values.

Overall, the findings support the conclusion that targeted, graph-aware recovery mechanisms represent a practical and low-overhead improvement path for existing decentralized multi-UAV exploration systems.

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APPENDIXES

Per-Run Raw Data Table

Run_ID	Environment	Configuration	Run	Sim_Time_s	Global_Plans
1	maze3	Baseline	1	2462	102
2	maze3	Baseline	2	2437	91
3	maze3	Baseline	3	2502	88
4	maze3	Baseline	4	2448	113
5	maze3	Baseline	5	2435	111
6	maze4	Baseline	1	678	64
7	maze4	Baseline	2	675	71
8	maze4	Baseline	3	817	62
9	maze4	Baseline	4	671	73
10	maze4	Baseline	5	654	87
11	testcity	Baseline	1	2098	0
12	testcity	Baseline	2	2031	0
13	testcity	Baseline	3	1996	0
14	testcity	Baseline	4	2063	0
15	testcity	Baseline	5	1993	0
16	maze3	IFER	1	532	210
17	maze3	IFER	2	509	221
18	maze3	IFER	3	487	208
19	maze3	IFER	4	514	215
20	maze3	IFER	5	505	213
21	maze4	IFER	1	458	55
22	maze4	IFER	2	441	52
23	maze4	IFER	3	469	57
24	maze4	IFER	4	453	54
25	maze4	IFER	5	456	52
26	testcity	IFER	1	2301	4
27	testcity	IFER	2	2214	3
28	testcity	IFER	3	2398	5
29	testcity	IFER	4	2329	3
30	testcity	IFER	5	2337	3

Appendix 2

IFER Implementation and Experiment Configuration Parameters

Category	Parameter	Value	Thesis Section
UAV Team	Number of UAVs	3	Section 4.1
	UAV platform model	Rotors-based simulated UAV (Gazebo)	Section 4.1
	Coordination type	Decentralized (no central controller)	Section 4.2
Simulation Environment	Simulator	Gazebo (ROS Noetic)	Section 4.1
	Test environments	testcity, maze3, maze4	Section 4.1
	Mapping method	Depth-based occupancy mapping	Section 4.1
	Baseline framework	GVP-MREP (unmodified stack)	Section 4.2
IFER Recovery Node	Implementation type	Auxiliary ROS node (parallel to baseline)	Section 4.3
	Exhaustion detection method	Trajectory topic silence monitoring	Section 3.4 / 4.3
	Silence threshold (exhaustion trigger)	21-22 seconds	Section 4.5
	Goal reassignment method	Dijkstra-based EROI search on MR-DTG	Section 3.4
	GVP-MREP stack modified?	No (overlay only)	Section 4.3
	Recovery target source	Shared MR-DTG node candidates	Section 3.4
	Post-recovery behavior	UAV resumes normal GVP-MREP planning cycle	Section 4.3
Experiment Configuration	Runs per environment (baseline)	5	Section 4.4
	Runs per environment (IFER)	5	Section 4.4
	Total runs	30	Section 4.4
	Stopping condition	Full stagnation or time limit reached	Section 4.5
	Data collection method	ROS bag recording + log files	Section 4.6
	Analysis pipeline	Offline CSV aggregation from rosbag/logs	Section 4.6